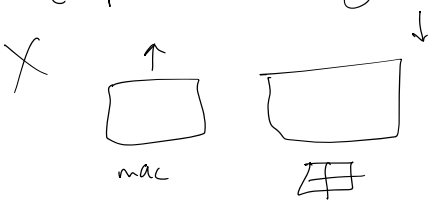
Lec32
30 March 2026 09:06 AM

Time complexity & space

Big input, less time
less memory

① Asymptotic notation & Experimental way

Experimental way

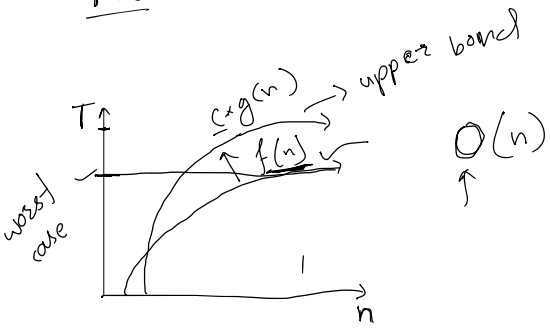


$i = 0$
$k = 20$
$n = 30$

worst case → 1 cr

best → 200 +

Average → 100



$O(n)$

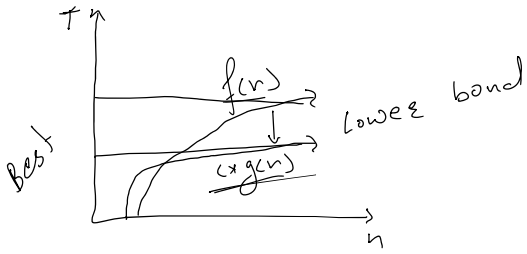
c = constant

$n = 10$

$f(n)$

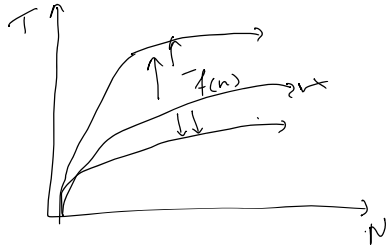
$\longrightarrow g(n)$

$f(n) <= c \times g(n)$



Ω

$f(n) >= c \cdot g(n)$



10000

$i = 1, 2, i = 3$ ... $i = 10$

→ for(i = 10; i>0; i++){

}

+
-
/
*
%

i = 1, 2, i = 3 ... i = 10

1 1 1 1

O(n)

i = 10, 9

1 1

10

(i = i i<=n; i * 2){

i = 1, i = 2, i = 4, 8, 16 → n

1 1 1 1

}

O(logn)

n²

for(i = 1; i<n; i++){

for(j = 1; j<n; j++){

}

}

i = 1, i = 2 ... n

n n ... n

n × n

n²

f(i = j; i<n; i++){ → n

for(j = 1; j<n; j*2){ → logn

}

}

n logn

→ log n²

n

+ -

/ *

L = 0

5

$j = 1$

1×2

$= 2$

2×2

$= 4$

4×2

8

$H = 10$

/ , *

log n

5

2

1

i

20

i = 1 , i = 2 , 3 , 4 , 5

1×1 2×2 3×3 4×4 5×5

1<=n 4<=n 16<=n 25<=n

```
for (i = 1; i * i <= n; i++) {
    System.out.println("hey");
}
```

$O(\sqrt{n})$

i = 1 , n<10

i = 10

10*

100

for (j = 1 ; j <= i*i ; j ++)

10